

Darrel Feiro Wihandi

✉ dfwihand@uwaterloo.ca — 🌐 Portfolio — 📄 Darrel Wihandi — 📄 DarrelFW321

Education

University of Waterloo

BSE Software Engineering Candidate

Expected Graduation: June 2029

Relevant Coursework: Data Structures & Data Management, Digital Computers, Database Management, Software Engineering Principles, Combinatorics, Compilers

Skills

Languages: C++, Python, Rust, HLSL, WGSL, TypeScript

Graphics & Compilers: SPIR-V, Vulkan, WebGPU, LLVM, CUDA, RenderDoc, CMake

Systems & Backend: Linux, Git, gRPC, PostgreSQL, Redis, AWS, Docker, Kubernetes

Experience

Software Engineering Intern | C++, Python, gRPC, PostgreSQL, Redis

Sept. 2026 – Present

Carta

- Built a **self-service enterprise SSO workflow** with C++ gRPC APIs for **configuration CRUD operations**, replacing staff-assisted setup and reducing estimated onboarding time from **1–2 hours to within 15 minutes**.
- Extended enterprise SSO integration with **SCIM provisioning** for user lifecycle and group membership changes, propagating identity updates across downstream services.
- Built a **Python mock identity provider** to simulate SSO authentication and validate SCIM provisioning, enabling automated integration testing of **access enforcement and portfolio transfer workflows**.

Graphics Compiler Contributor | C++, Slang, SPIR-V, WGSL

Feb. 2025 – Present

NVIDIA Slang – Open Source

- Extended Slang's **SPIR-V and WGSL backends** to generate texture-dimension queries for 2D and array textures, handling **mip-level arguments and target-specific result types**.
- Diagnosed **cross-backend code-generation inconsistencies** by inspecting emitted shader code, adding regression tests to validate texture-query behavior across supported targets.

Machine Learning Systems Engineer | Python, OpenTelemetry, SQLite

May 2026 – Oct. 2026

Wat.AI – TRACE

- Built **TRACE**, a Python trace-analysis framework using **deterministic state validation** to identify failure origins in multi-step AI-agent executions and generate reproducible regression cases.
- Automated **500+ evaluation runs** across **30+ adversarial scenarios**, integrating regression replay into CI to detect recurring failures and verify fixes preserve valid agent behavior.

AI Software Engineer | Python, Azure OpenAI, FastAPI

Jan. 2026 – May 2026

Rogers Communications

- Developed a **metadata enrichment application** integrating Collibra APIs, Azure OpenAI, and business-glossary retrieval, with validation and human review; reduced authoring time from **6 to 2 minutes per asset**.

Software Engineer | C++, CUDA, Python, AWS EKS

May 2025 – Aug. 2025

Eon Media

- Implemented C++/CUDA kernels for **fused frame resizing and color conversion** in a real-time sports analytics pipeline, reducing preprocessing latency by **30% versus CPU OpenCV** ahead of YOLO inference.
- Deployed **GPU-accelerated video processing services on AWS EKS** with GPU resource configuration, health checks, and S3 integration; automated CI/CD across EKS and EC2, reducing deployment overhead by **50%**.

Projects

Flux 🌀 | C++, SPIR-V, WebGPU, WebAssembly

Feb. 2026 – Mar. 2026

Shader Language, Compiler, and Interactive GPU Playground

- Designed a **statically typed shader language in C++**, implementing parsing, type checking, and a custom intermediate representation for shader programs.
- Implemented **IR optimization passes** and compiler backends generating **SPIR-V and WGSL** shader programs.
- Built an **interactive WebGPU playground** for editing and previewing shaders, with live diagnostics and a WebAssembly visualizer showing how source changes affect compiled output.

Nimbus ☁ | C++, Vulkan, GLSL, HLSL, RenderDoc

Mar. 2026 – May 2026

Real-Time Volumetric Renderer

- Built a **Vulkan-based volumetric renderer** with GPU ray-marched clouds, atmospheric scattering, dynamic lighting, and temporal reconstruction.
- Reduced **1080p GPU frame time from 12.19 ms to 2.38 ms (5.1× speedup)** through adaptive ray marching, early termination, temporal reconstruction, and GPU profiling.